DATABASES

Backup & Recovery Visual Guide

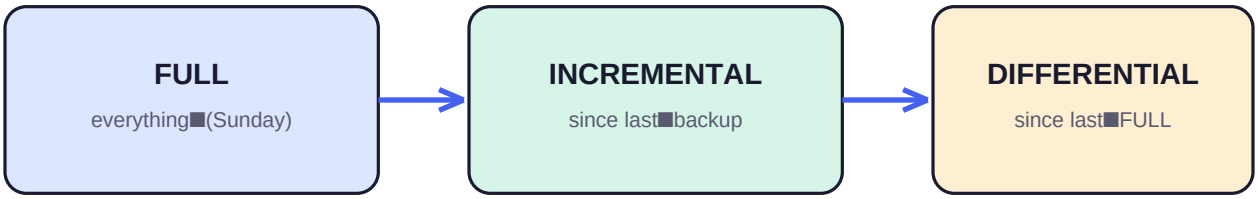
Full, incremental, PITR --
backups you can actually restore from.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Backup Types

Three flavors, one timeline.

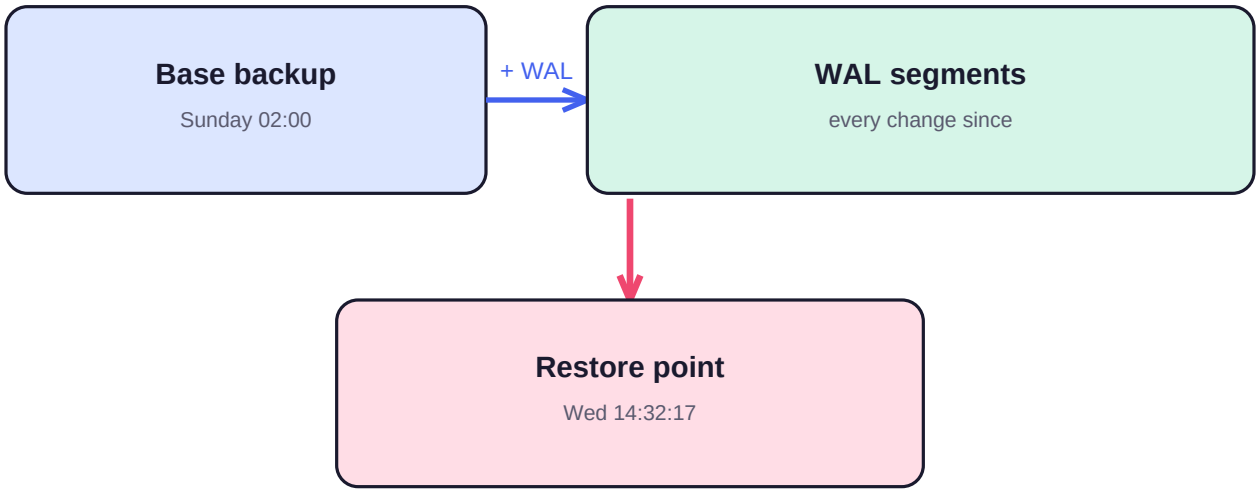


- Full: slowest, simplest restore -- one file, no chain to break.
- Incremental: fast backups, but restore replays the whole chain -- one corrupt link kills it.
- Differential: middle ground -- one full + one diff covers any day since.

STEP 2/6

PITR with WAL Archiving

Restore to ANY second, not just backup time.



- Archive every WAL segment off the primary (archive_command) -- gaps mean unrecoverable time.
- Recovery replays: base backup + WAL up to your target timestamp.
- Test the archive stream continuously -- a silent archiving failure voids all PITR promises.

STEP 3/6

Backup Storage: 3-2-1

Where backups live matters as much as taking them.

3 copies

primary + backup + offsite

2 media types

disk + object storage (S3-class)

1 offsite

different region -- survives the datacenter fire

Encrypt backups at rest. A stolen backup is a stolen database -- keys managed separately.

STEP 4/6

Restore Testing Discipline

An untested backup is a rumor.

1. Automate

nightly restore into a scratch environment

2. Verify

row counts, checksums, app smoke tests

3. Time it

measure real RTO -- hope is not a metric

4. Document

runbook anyone can follow at 3 AM

STEP 5/6

RTO & RPO

Two numbers that define your whole strategy.

RTO

how LONG to restore (downtime budget)

RPO

how MUCH data to lose (backup frequency)

- RPO 5 min + RTO 1 hr demands: continuous WAL archiving + rehearsed restores.
- Tighter targets cost real money -- get the business to sign the numbers.
- Your ACTUAL RTO is whatever the last restore drill measured, not the SLA slide.

STEP ✓/✓

Key Takeaways

Backup & Recovery -- Remember

- Full backups restore simply; incremental chains are fragile -- pick deliberately.
- PITR = base backup + continuous WAL archiving -- restores to any second, if the stream is healthy.
- 3-2-1: three copies, two media, one offsite -- encrypted, keys separate.
- Restore on a schedule: automated, verified, timed, documented -- or it doesn't count.
- RTO/RPO are business decisions -- but only the last drill proves you meet them.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev